
Session 01 (AIML) - Assignment Questions

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Branch: **Artificial Intelligence and Machine Learning (AIML)**

Domain: **AIML (Artificial Intelligence and Machine Learning)**

Q1. Variables & Naming Rules

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  JUPYTER

● PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Full Name: Vaishnavi Praveen Rathi
Age: 17
Height: 1.25 meters
Is Student: True

Data type of full_name: <class 'str'>
Data type of age: <class 'int'>
Data type of height: <class 'float'>
Data type of is_student: <class 'bool'>
PS C:\Users\rathi_pu888d0\Desktop\ITR> █
```

Program Code:

```
""" ASSIGNMENT 1 """
```

```
'''
```

Q1. (Variables & Naming Rules)

Create variables to store the following information about yourself:

- Your full name (string)
- Your age (integer)
- Your height in meters (float)
- Whether you are a student (boolean)

Print all four variables. Then explain in comments why you chose those variable names and check their data types using type()

```
'''
```

```
# Initialize variables with information about myself
```

```
full_name = "Vaishnavi Praveen Rath"i
```

```
age = 17
```

```
height = 1.25
```

```
is_student = True
```

```
# Print variable values
```

```
print("Full Name:", full_name)
```

```
print("Age:", age)
```

```
print("Height:", height, "meters")
```

```
print("Is Student:", is_student)
```

```
print("\n") # Blank line
```

```
# Check data types
```

```
print("Data type of full_name:", type(full_name))
```

```
print("Data type of age:", type(age))
```

```
print("Data type of height:", type(height))
```

```
print("Data type of is_student:", type(is_student))
```

```
"""Explanation in Comments"""
```

```
# The variable 'full_name' is a string that holds my full name.
```

```
# The variable 'age' is an integer that represents my age in years.
```

```
# The variable 'height' is a float that represents my height in meters.
```

```
# The variable 'is_Student' is a boolean that indicates whether I am a student or not
```

Q2. Data Types

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  JUPYTER

● PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Book Name: Python Programming
Data Type of book_name: <class 'str'>
Pages: 250
Data Type of pages: <class 'int'>
Price: 499.99
Data Type of price: <class 'float'>
Is Available: True
Data Type of is_available: <class 'bool'>

Pages converted to float: 250.0
Data Type: <class 'float'>
Price converted to integer: 499
Data Type: <class 'int'>
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

```
""" ASSIGNMENT 1 """

"""
Q2. (Data Types)
Write a Python program that:
a) Creates one variable of each core data type: str, int, float, and bool.
b) Prints each variable along with its data type using type().
c) Try converting the integer to float and float to integer. Print the results and
explain the output.
"""

# Creating variables of different data types
book_name = "Python Programming"
pages = 250
price = 499.99
is_available = True
```

```
# Printing values and their data types
print("Book Name:", book_name)
print("Data Type of book_name:", type(book_name))
print("Pages:", pages)
print("Data Type of pages:", type(pages))
print("Price:", price)
print("Data Type of price:", type(price))
print("Is Available:", is_available)
print("Data Type of is_available:", type(is_available))

print()
```

Type Conversion

```
pages_float = float(pages)
price_int = int(price)

print("Pages converted to float:", pages_float)
print("Data Type:", type(pages_float))
print("Price converted to integer:", price_int)
print("Data Type:", type(price_int))
```

"Explanation in Comments"

```
# float(pages) converts the integer value into a float.
# int(price) converts the float value into an integer.
# The decimal part (.99) is removed during conversion.
```

Q3. Operators - Arithmetic

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  JUPYTER

PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Addition: 19
Subtraction: 11
Multiplication: 60
Division: 3.75
Floor Division: 3
Modulo: 3
Exponentiation: 50625
PS C:\Users\rathi_pu888d0\Desktop\ITR> █
```

Program Code:

```
""" ASSIGNMENT 1 """
```

```
'''
```

```
Q3. (Operators - Arithmetic)
```

Write a program that takes two numbers $a = 15$ and $b = 4$ and prints the results of: Addition, Subtraction, Multiplication, Division, Floor Division, Modulo, and Exponentiation.

Add comments explaining the difference between $/$ and $//$ operators with an example.

```
'''
```

```
# Initialize variables
```

```
a = 15
```

```
b = 4
```

Arithmetic Operations

```
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Floor Division:", a // b)
print("Modulo:", a % b)
print("Exponentiation:", a ** b)
```

"Explanation in Comments"

/ performs normal division and returns the exact quotient.

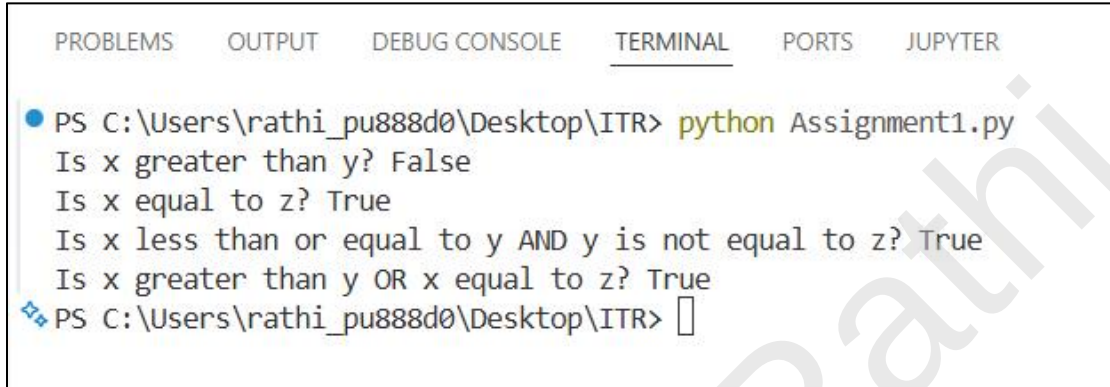
Example: $15 / 4 = 3.75$

// performs floor division and returns only the integer part.

Example: $15 // 4 = 3$

Q4. Operators - Comparison & Logical

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Is x greater than y? False
Is x equal to z? True
Is x less than or equal to y AND y is not equal to z? True
Is x greater than y OR x equal to z? True
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

```
""" ASSIGNMENT 1 """
```

```
'''
```

Q4. (Operators - Comparison & Logical)

Given:

x = 25

y = 30

z = 25

Write expressions using comparison and logical operators to check:

1. Is x greater than y?
2. Is x equal to z?
3. Is x less than or equal to y AND y is not equal to z?
4. Is x greater than y OR x equal to z?

Print all results with descriptive messages.

```
'''
```

```
# Initialize variables
```

```
x = 25
```

```
y = 30
```

```
z = 25
```

Comparison and Logical Operations

```
print("Is x greater than y?", x > y)
```

```
print("Is x equal to z?", x == z)
```

```
print("Is x less than or equal to y AND y is not equal to z?",  
      x <= y and y != z)
```

```
print("Is x greater than y OR x equal to z?",  
      x > y or x == z)
```

Q5. Input/Output

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

● PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Enter your name: Vaishnavi
Enter your birth year: 2008

Name: Vaishnavi
Age: 18
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

""" ASSIGNMENT 1 """

'''

Q5. (Input/Output)

Write a program that asks the user to input:

- Their name
- Their birth year

Then calculate and print their current age
(Assume current year is 2026).

Use proper formatting and add comments explaining
why you need to convert the input.

'''

Taking input from the user

```
name = input("Enter your name: ")
birth_year = int(input("Enter your birth year: "))
```

We convert the input to int because input() returns data as a string.
Mathematical operations cannot be performed directly on strings.

```
current_year = 2026  
age = current_year - birth_year
```

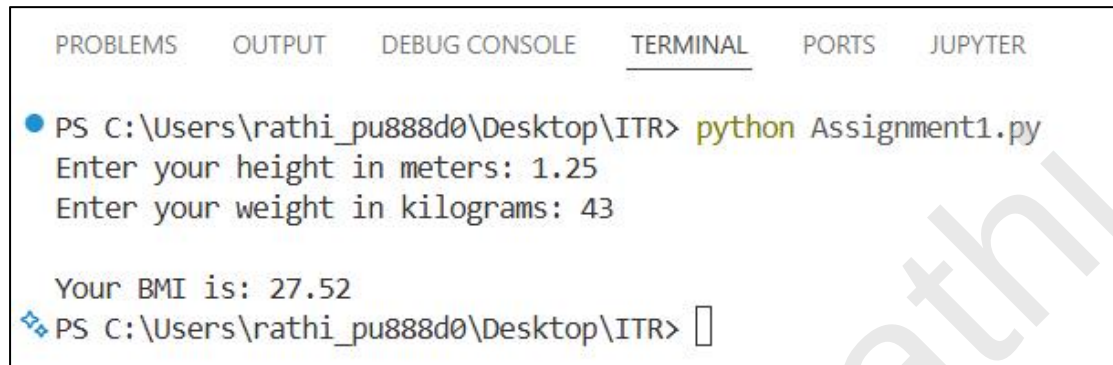
```
# Displaying the result
```

```
print()  
print("Name:", name)  
print("Age:", age)
```

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Q6. Combined - Variables, Operators & Input

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Enter your height in meters: 1.25
Enter your weight in kilograms: 43

Your BMI is: 27.52
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

```
""" ASSIGNMENT 1 """
```

```
'''
```

Q6. (Combined - Variables, Operators & Input)

Create a simple BMI Calculator:

Take height (in meters) and weight (in kg) as input from the user.

Calculate BMI using the formula:

$BMI = weight / (height ** 2)$

Print the BMI rounded to 2 decimal places.

Include appropriate comments.

```
'''
```

```
# Taking input from the user
```

```
height = float(input("Enter your height in meters: "))
```

```
weight = float(input("Enter your weight in kilograms: "))
```

```
# Calculating BMI using the given formula
```

```
bmi = weight / (height ** 2)
```

```
# Displaying the BMI rounded to 2 decimal places
```

```
print()
```

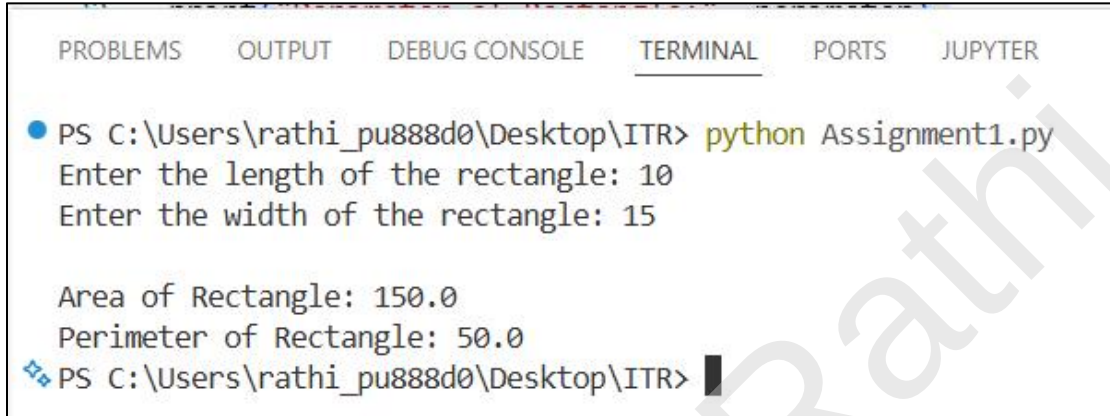
```
print("Your BMI is:", round(bmi, 2))
```

```
# Explanation:  
# float() is used because height and weight may contain decimal values.  
# height ** 2 calculates the square of the height.  
# round(bmi, 2) rounds the BMI value to 2 decimal places.
```

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Q7. Comments

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Enter the length of the rectangle: 10
Enter the width of the rectangle: 15

Area of Rectangle: 150.0
Perimeter of Rectangle: 50.0
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

```
""" ASSIGNMENT 1 """
```

```
'''
```

Q7. (Comments)

Write a small program that calculates the area and perimeter of a rectangle.
Take length and width as input.
Use at least 5 meaningful comments (single-line and one multi-line comment).
Follow good commenting practices (explain why, not what for obvious lines)

```
'''
```

```
# Length and width may contain decimal values, so float() is used.
```

```
length = float(input("Enter the length of the rectangle: "))
```

```
width = float(input("Enter the width of the rectangle: "))
```

```
'''
```

Area helps determine the amount of space inside the rectangle,
while perimeter gives the total distance around its boundary.
Both are commonly used in geometry and real-world measurements.

```
'''
```

```
# Area is needed when measuring the surface covered by a shape.
```

```
area = length * width
```

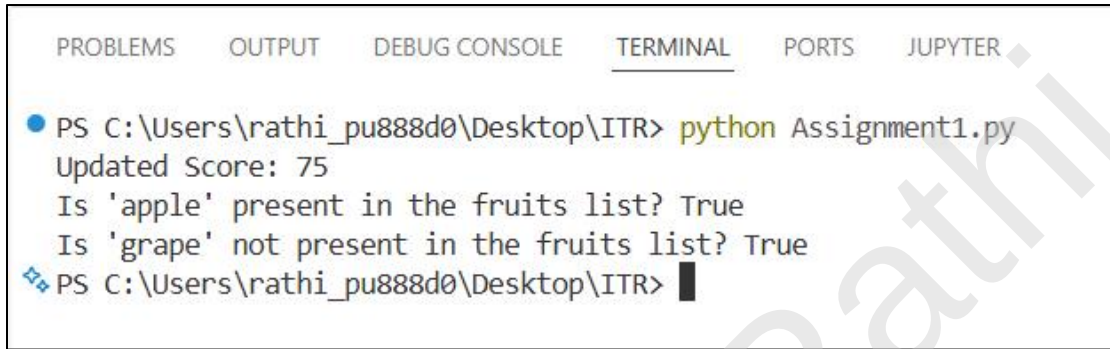
```
# Perimeter is useful for finding the total boundary length.  
perimeter = 2 * (length + width)
```

```
# Clear labels make the output easier for users to understand.  
print()  
print("Area of Rectangle:", area)  
print("Perimeter of Rectangle:", perimeter)
```

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Q8. Operators - Assignment & Membership

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  JUPYTER

PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Updated Score: 75
Is 'apple' present in the fruits list? True
Is 'grape' not present in the fruits list? True
PS C:\Users\rathi_pu888d0\Desktop\ITR> █
```

Program Code:

```
""" ASSIGNMENT 1 """

"""
Q8. (Operators - Assignment & Membership)

fruits = ["mango", "apple", "banana", "cherry"]
score = 50

Perform the following operations:
1. Add 25 to score using assignment operator (+=).
2. Check if "apple" is in the fruits list.
3. Check if "grape" is not in the list.

Print all results with clear messages.
"""

# Initialize variables

fruits = ["mango", "apple", "banana", "cherry"]
score = 50

# Using assignment operator to increase the score

score += 25
```

Checking membership in the list

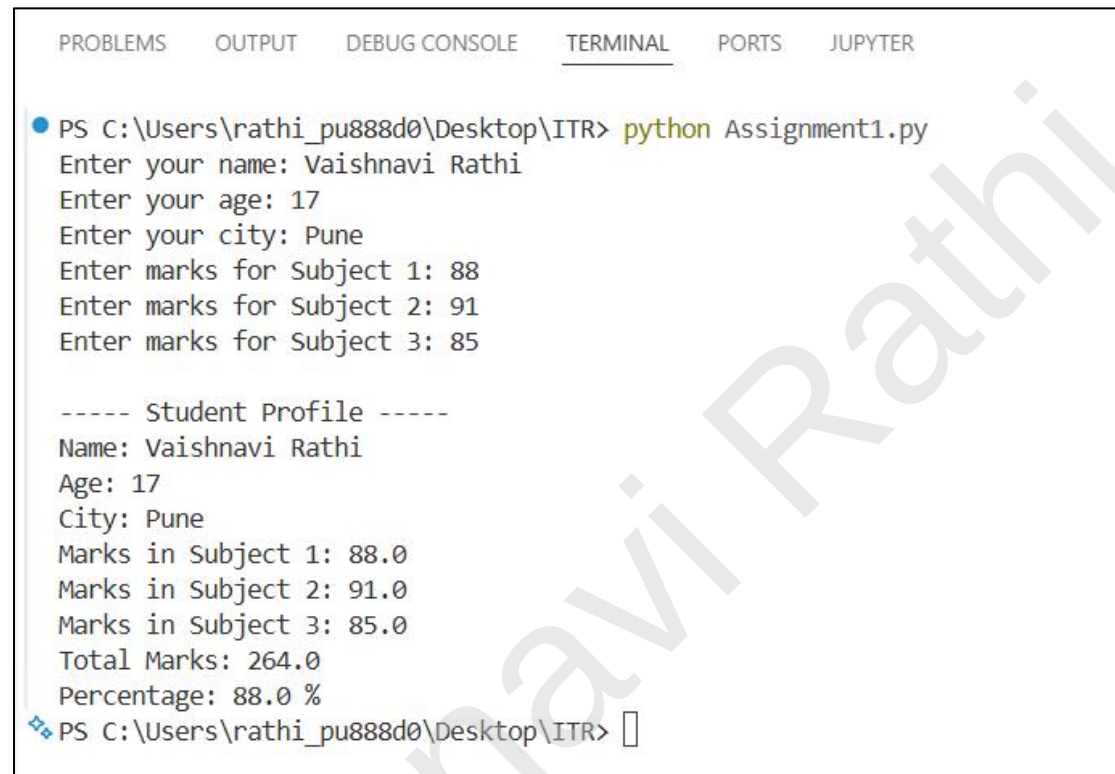
```
apple_present = "apple" in fruits  
grape_absent = "grape" not in fruits
```

Displaying results

```
print("Updated Score:", score)  
print("Is 'apple' present in the fruits list?", apple_present)  
print("Is 'grape' not present in the fruits list?", grape_absent)
```

Q9. Mini Project - All Concepts

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER
PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
Enter your name: Vaishnavi Rathi
Enter your age: 17
Enter your city: Pune
Enter marks for Subject 1: 88
Enter marks for Subject 2: 91
Enter marks for Subject 3: 85

----- Student Profile -----
Name: Vaishnavi Rathi
Age: 17
City: Pune
Marks in Subject 1: 88.0
Marks in Subject 2: 91.0
Marks in Subject 3: 85.0
Total Marks: 264.0
Percentage: 88.0 %
PS C:\Users\rathi_pu888d0\Desktop\ITR> 
```

Program Code:

```
""" ASSIGNMENT 1 """
```

```
'''
```

Q9. (Mini Project - All Concepts)

Write a Student Profile Generator program that:

- Takes input: name, age, city, and marks in 3 subjects.
- Calculates total marks and percentage.
- Prints a formatted profile with all information.
- Use meaningful variable names, proper comments, and different data types.

```
'''
```

```
# Taking student details as input
```

```
student_name = input("Enter your name: ")
student_age = int(input("Enter your age: "))
student_city = input("Enter your city: ")
```

Taking marks for three subjects

```
subject1_marks = float(input("Enter marks for Subject 1: "))
subject2_marks = float(input("Enter marks for Subject 2: "))
subject3_marks = float(input("Enter marks for Subject 3: "))
```

Calculating total marks obtained

```
total_marks = subject1_marks + subject2_marks + subject3_marks
```

Assuming each subject is out of 100 marks

```
percentage = (total_marks / 300) * 100
```

Displaying the student profile in a formatted way

```
print("\n----- Student Profile -----")
print("Name:", student_name)
print("Age:", student_age)
print("City:", student_city)
print("Marks in Subject 1:", subject1_marks)
print("Marks in Subject 2:", subject2_marks)
print("Marks in Subject 3:", subject3_marks)
print("Total Marks:", total_marks)
print(f"Percentage: {percentage:.2f}%")
```

Q10. Debugging & Understanding

```
Name = "Alice"
age 20
city = Amsterdam
print("My name is" + Name)
print("I am" age "years old")
print("I live in " city)
# Check if age is greater than 18 and print message
print("Adult:" age > 18)
```

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

PS C:\Users\rathi_pu888d0\Desktop\ITR> python Assignment1.py
My name is Alice
I am 20 years old
I live in Amsterdam
Adult: True
PS C:\Users\rathi_pu888d0\Desktop\ITR> █
```

Fixed Program Code:

```
""" ASSIGNMENT 1 """
```

```
'''
```

Q10. (Debugging & Understanding)

The following code has some issues.

Correct it and add proper comments explaining the fixes.

```
'''
```

```
Name = "Alice"
```

```
age = 20 # Added assignment operator (=)
```

```
city = "Amsterdam" # Added quotation marks
```

```
print("My name is " + Name)    # We can use + to concatenate strings but we need  
String value for Name  
print("I am", age, "years old") # Used commas to print variable  
print("I live in", city)       # Used comma before variable  
  
print("Adult:", age > 18)      # Added comma to print the result
```

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